

# Linear Regression Models

Materials from Philippe Rigollet's Lecture Notes of High Dimensional Statistics

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## Definition 1 (Sub-Gaussian Variable)

A random variable  $X \in \mathbb{R}$  is said to be sub-Gaussian with variance proxy  $\sigma^2$  if  $\mathbb{E}[X] = 0$  and its moment generating function satisfies

$$\mathbb{E}[\exp(sX)] \leq \exp\left(\frac{\sigma^2 s^2}{2}\right), \quad \forall s \in \mathbb{R}.$$

In this case we write  $X \sim \text{subG}(\sigma^2)$ .

## Definition 2 (Sub-Gaussian Vector)

A random vector  $X \in \mathbb{R}^d$  is said to be sub-Gaussian with variance proxy  $\sigma^2$  if  $\mathbb{E}[X] = 0$  and  $u^T X$  is sub-Gaussian with variance proxy  $\sigma^2$  for any unit vector  $u \in S^{d-1}$ . In this case we write  $X \sim \text{subG}_d(\sigma^2)$ .

## Definition 3 (Sub-Gaussian Matrix)

A random vector  $X \in \mathbb{R}^{d \times T}$  is said to be sub-Gaussian with variance proxy  $\sigma^2$  if  $\mathbb{E}[X] = 0$  and  $u^T X v$  is sub-Gaussian with variance proxy  $\sigma^2$  for any unit vector  $u \in S^{d-1}$ ,  $v \in S^{d-1}$ . In this case we write  $X \sim \text{subG}_{d \times T}(\sigma^2)$ .

## Theorem 4

Let  $X \in \mathbb{R}^d$  be a sub-Gaussian random vector with variance proxy  $\sigma^2$  and  $\mathcal{B}_2 = \{\theta \in \mathbb{R}^d : \|\theta\|_2 \leq 1\}$ . Then

$$\mathbb{E}[\max_{\theta \in \mathcal{B}_2} \theta^T X] = \mathbb{E}[\max_{\theta \in \mathcal{B}_2} |\theta^T X|] \leq 4\sigma\sqrt{d} \quad \text{and} \quad \mathbb{P}(\max_{\theta \in \mathcal{B}_2} \theta^T X > t) \leq 6^d e^{-\frac{t^2}{8\sigma^2}},$$

for any  $t > 0$ . Moreover, for any  $\delta > 0$ , with probability  $1 - \delta$ , it holds that

$$\max_{\theta \in \mathcal{B}_2} \theta^T X = \max_{\theta \in \mathcal{B}_2} |\theta^T X| \leq 4\sigma\sqrt{d} + 2\sigma\sqrt{2\log(1/\delta)}.$$

## Preliminaries

Let  $\mathcal{V}(P)$  denote a set of finite number of vertices, a convex polytope  $P$  is defined as  
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### Theorem 5

Let  $P$  be a polytope with  $N$  vertices  $v^{(1)}, \dots, v^{(N)} \in \mathbb{R}^d$  and let  $X \in \mathbb{R}^d$  be a random vector such that,  $[v^{(i)}]^T X$ ,  $i = 1, \dots, N$  are sub-Gaussian random variables with variance proxy  $\sigma^2$ . Then

$$\mathbb{E}[\max_{\theta \in P} \theta^T X] \leq \sigma \sqrt{2 \log N}, \quad \text{and} \quad \mathbb{E}[\max_{\theta \in P} |\theta^T X|] \leq \sigma \sqrt{2 \log(2N)}.$$

Moreover, for any  $t > 0$ ,

$$\mathbb{P} \left( \max_{\theta \in P} \theta^T X > t \right) \leq N e^{-\frac{t^2}{2\sigma^2}}, \quad \text{and} \quad \mathbb{P} \left( \max_{\theta \in P} |\theta^T X| > t \right) \leq 2N e^{-\frac{t^2}{2\sigma^2}}.$$



# Fixed Design Linear Regression

Consider the following model:

$$Y_i = f(x_i) + \epsilon_i, \quad i = 1, 2, \dots, n,$$

where  $\epsilon = (\epsilon_1, \dots, \epsilon_n)^T$  is sub-Gaussian with variance proxy  $\sigma^2$  and  $\mathbb{E}[\epsilon] = 0$ . We assume that  $x \in \mathbb{R}^d$  and  $f(x) = x^T \theta^*$  for some unknown  $\theta^*$ .

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In contrast to the random design where  $x_1, \dots, x_n$  are **stochastic**, the fixed design assumes that  $x_1, \dots, x_n$  are **deterministic**. In such setting, we observe  $\mu^* = (f(x_1), \dots, f(x_n))^T \in \mathbb{R}^n$  and  $\epsilon = (\epsilon_1, \dots, \epsilon_n)^T \in \mathbb{R}^n$  is sub-Gaussian with variance proxy  $\sigma^2$ .

# Fixed Design Linear Regression

We focus on the *Mean Squared Error*(MSE) as a measure of performance. It is defined by

$$\text{MSE}(\hat{f}_n) = \frac{1}{n} \sum_{i=1}^n \left( \hat{f}_n(x_i) - f(x_i) \right)^2.$$

Or equivalently,

$$\text{MSE}(\hat{\mu}) = \frac{1}{n} \sum_{i=1}^n (\hat{\mu}_i - \mu_i^*)^2 = \|\hat{\mu} - \mu^*\|_2^2$$

# Fixed Design Linear Regression

Let the design vectors  $x_1, \dots, x_n \in \mathbb{R}^d$  store in a  $n \times d$  matrix  $\mathbb{X}$ , whose  $j$ th row is given by  $x_j^T$ . The linear regression model can be written in the matrix form:

$$Y = \mathbb{X}\theta^* + \epsilon, \quad (\text{Linear Model})$$

where  $Y = (Y_1, \dots, Y_n)^T$  and  $\epsilon = (\epsilon_1, \dots, \epsilon_n)$ . Moreover,

$$\text{MSE}(\mathbb{X}\hat{\theta}) = \frac{1}{n} \|\mathbb{X}(\hat{\theta} - \theta^*)\|_2^2 = (\hat{\theta} - \theta^*)^T \frac{\mathbb{X}^T \mathbb{X}}{n} (\hat{\theta} - \theta^*)$$

# Unconstrained Least Squares Estimator

## Definition 6 (Least Squares Estimator)

Let the least squares estimator  $\hat{\theta}^{LS}$  be any vector such that

$$\hat{\theta}^{LS} \in \arg \min_{\theta \in \mathbb{R}^d} \|Y - \mathbb{X}\theta\|_2^2.$$

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We also call  $\hat{\mu}^{LS} = \mathbb{X}\hat{\theta}^{LS}$  least squares estimator. By definition of the least squares estimator,  $\hat{\mu}^{LS}$  can be regarded as a projection of  $Y$  onto the space spans by the columns of  $\mathbb{X}$ .

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It is known that the least squares estimators of  $\theta^*$  and  $\mu^* = \mathbb{X}\theta^*$  are maximum likelihood estimators when  $\epsilon \sim \mathcal{N}(0, \sigma^2 I_n)$ .

# Unconstrained Least Squares Estimator

## Proposition 1

The least squares estimator  $\hat{\mu}^{LS} = \mathbb{X}\hat{\theta}^{LS}$  satisfies

$$\mathbb{X}^T \hat{\mu}^{LS} = \mathbb{X}^T Y.$$

Moreover,  $\hat{\theta}^{LS}$  can be chosen to be

$$\hat{\theta}^{LS} = (\mathbb{X}^T \mathbb{X})^\dagger \mathbb{X}^T Y,$$

where  $(\mathbb{X}^T \mathbb{X})^\dagger$  denotes the Moore-Penrose pseudoinverse of  $\mathbb{X}^T \mathbb{X}$ .



## Proof of Proposition 1.

Let  $h(\theta) = \|Y - \mathbb{X}\theta\|_2^2$ , which is convex and differentiable over  $\theta \in \mathbb{R}^d$ , so any of its minima  $\hat{\theta}^{LS}$  satisfies

$$\nabla_{\theta} h(\hat{\theta}^{LS}) = 2\mathbb{X}^T(Y - \mathbb{X}\hat{\theta}^{LS}) = 0.$$

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That is

$$\mathbb{X}^T\mathbb{X}\hat{\theta}^{LS} = \mathbb{X}^TY.$$

It concludes the proof of the first statement. The second statement follows from the definition of the Moore-Penrose pseudoinverse.  $\square$

# Unconstrained Least Squares Estimator

## Theorem 7

Assume that (Linear Model) holds where  $\epsilon \sim \text{subG}_n(\sigma^2)$ . Then the least squares estimator  $\hat{\theta}^{LS}$  satisfies

$$\mathbb{E} \left[ \text{MSE}(\mathbb{X} \hat{\theta}^{LS}) \right] = \frac{1}{n} \mathbb{E} \|\mathbb{X} \hat{\theta}^{LS} - \mathbb{X} \theta^*\|_2^2 \lesssim \sigma^2 \frac{r}{n}.$$

where  $r = \text{rank}(\mathbb{X}^T \mathbb{X})$ . Moreover, for any  $\delta > 0$ , with probability  $1 - \delta$ , it holds

$$\text{MSE}(\mathbb{X} \hat{\theta}^{LS}) \lesssim \sigma^2 \frac{r + \log(1/\delta)}{n}.$$

## Proof of Theorem 7.

By definition of  $\hat{\theta}^{LS}$ ,

$$\|Y - \mathbb{X}\hat{\theta}^{LS}\|_2^2 \leq \|Y - \mathbb{X}\theta^*\|_2^2 = \|\epsilon\|_2^2.$$

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Moreover,

$$\|Y - \mathbb{X}\hat{\theta}^{LS}\|_2^2 = \|\mathbb{X}\theta^* + \epsilon - \mathbb{X}\hat{\theta}^{LS}\|_2^2 = \|\mathbb{X}\theta^* - \mathbb{X}\hat{\theta}^{LS}\|_2^2 - 2\epsilon^T \mathbb{X}(\hat{\theta}^{LS} - \theta^*) + \|\epsilon\|_2^2$$

Therefore, we get

$$\|\mathbb{X}\theta^* - \mathbb{X}\hat{\theta}^{LS}\|_2^2 \leq 2\epsilon^T \mathbb{X}(\hat{\theta}^{LS} - \theta^*) = 2\|\mathbb{X}\theta^* - \mathbb{X}\hat{\theta}^{LS}\|_2 \frac{\epsilon^T \mathbb{X}(\hat{\theta}^{LS} - \theta^*)}{\|\mathbb{X}\theta^* - \mathbb{X}\hat{\theta}^{LS}\|_2}$$

□

## Proof of Theorem 7.

As  $\hat{\theta}^{LS}$  depends on  $\epsilon$ , it is difficult to control the term

$$\frac{\epsilon^T \mathbb{X}(\hat{\theta}^{LS} - \theta^*)}{\|\mathbb{X}\theta^* - \mathbb{X}\hat{\theta}^{LS}\|_2}$$

To remove this dependency, we apply a technique "sup-out"  $\hat{\theta}^{LS}$

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Let  $\Phi = [\varphi_1, \dots, \varphi_r] \in \mathbb{R}^{n \times r}$  be an orthonormal basis of the **range** of  $\mathbb{X}$ . Hence, there exists  $\nu \in \mathbb{R}^r$  such that  $\mathbb{X}(\hat{\theta}^{LS} - \theta^*) = \Phi\nu$ .

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$$\frac{\epsilon^T \mathbb{X}(\hat{\theta}^{LS} - \theta^*)}{\|\mathbb{X}\theta^* - \mathbb{X}\hat{\theta}^{LS}\|_2} = \frac{\epsilon^T \Phi\nu}{\|\Phi\nu\|_2} = \frac{\epsilon^T \Phi\nu}{\|\nu\|_2} = \frac{\tilde{\epsilon}^T \nu}{\|\nu\|_2} \leq \sup_{u \in \mathcal{B}_2} \tilde{\epsilon}^T u.$$

where  $\mathcal{B}_2$  is the unit ball of  $\mathbb{R}^r$  and  $\tilde{\epsilon} = \Phi^T \epsilon$ .



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$$\|\mathbb{X}\theta^* - \mathbb{X}\hat{\theta}^{LS}\|_2^2 \leq 4 \sup_{u \in \mathcal{B}_2} (\tilde{\epsilon}^T u)^2.$$

## Proof of Theorem 7.

For any  $u \in S^{r-1}$ , it holds  $\|\Phi u\|_2^2 = u^T \Phi^T \Phi u = u^T u = 1$ , which implies  $\Phi u \in S^{r-1}$ .

As  $\epsilon \sim \text{subG}_n(\sigma^2)$ , by definition we have

$$\mathbb{E} \left[ e^{s \epsilon^T \Phi u} \right] \leq e^{\frac{s^2 \sigma^2}{2}}, \quad \forall s \in \mathbb{R}.$$

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$$\mathbb{E} \left[ e^{s \tilde{\epsilon}^T u} \right] = \mathbb{E} \left[ e^{s \epsilon^T \Phi u} \right] \leq e^{\frac{s^2 \sigma^2}{2}}, \quad \forall s \in \mathbb{R}.$$

Therefore,  $\tilde{\epsilon} \sim \text{subG}_r(\sigma^2)$  and  $\tilde{\epsilon}_i \sim \text{subG}(\sigma^2)$

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$$\mathbb{E} \left[ \sup_{u \in \mathcal{B}_2} (\tilde{\epsilon}^T u)^2 \right] = \mathbb{E} \|\tilde{\epsilon}\|_2^2 = \sum_{i=1}^r \mathbb{E} [\tilde{\epsilon}_i^2]$$

where the last inequality comes from the property of sub Gaussian variable that  $\mathbb{E}[X^2] \leq 4\sigma^2$ .

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Putting together, we conclude the proof the first statement. □

## Proof of Theorem 7.

By the Theorem 4, with probability at least  $1 - \delta$ , it holds that

$$\sup_{u \in \mathcal{B}_2} (\tilde{\epsilon}^T u)^2 \leq \left( 4\sigma\sqrt{r} + 2\sigma\sqrt{2\log(1/\delta)} \right)^2 \leq 32\sigma^2 r + 16\sigma^2 \log(1/\delta)$$

where the last inequality comes from the fact that  $(a + b)^2 \leq 2a^2 + 2b^2$ . □

If  $d \leq n$  and  $B := \frac{\mathbb{X}^T \mathbb{X}}{n}$  has rank  $d$ , then we have

$$\|\hat{\theta}^{LS} - \theta^*\|_2^2 \leq \frac{\text{MSE}(\mathbb{X}\hat{\theta}^{LS})}{\lambda_{\min}(B)}, \quad (\lambda_{\min}(B) > 0 \text{ by assumption}).$$

and we can use Theorem 7 to bound  $\|\hat{\theta}^{LS} - \theta^*\|_2^2$  directly.

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For a  $d \times d$  symmetric matrix  $B$ , it has eigen-decomposition  $U\Sigma U^T$  where  $U^T U = U U^T = I$ . Any vector  $u \in S^{d-1}$  can be expressed by  $U\alpha$  for some  $\alpha \in \mathbb{R}^d$  satisfied  $\|\alpha\|_2 = 1$ .



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Hence, for any  $u \in S^{d-1}$ ,

$$u^T B u = (U\alpha)^T U \Sigma U^T (U\alpha) = \alpha^T \Sigma \alpha = \sum_{i=1}^d \lambda_i(B) \alpha_i^2 \geq \lambda_{\min}(B) \sum_{i=1}^n \alpha_i^2 = \lambda_{\min}(B).$$

## Constrained Least Squares Estimator

Let  $K \subset \mathbb{R}^d$  be a symmetric convex set. If we know a priori that  $\theta^* \in K$ , we may prefer a constrained least squares estimator  $\hat{\theta}_K^{LS}$  defined by

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And similarly,

$$\|\mathbb{X}\hat{\theta}_K^{LS} - \mathbb{X}\theta^*\|_2^2 \leq 2\epsilon^T \mathbb{X}(\hat{\theta}_K^{LS} - \theta^*) \leq 2 \sup_{\theta \in K-K} (\epsilon^T \mathbb{X}\theta)$$

where  $K - K \triangleq \{x - y : x, y \in K\}$ .

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$$2 \sup_{\theta \in K-K} (\epsilon^T \mathbb{X}\theta) = 4 \underbrace{\sup_{v \in \mathbb{X}K} (\epsilon^T v)}_{\text{a measure of the size of } \mathbb{X}K},$$

where  $\mathbb{X}K \triangleq \{\mathbb{X}\theta : \theta \in K\} \subset \mathbb{R}^n$ .

## $\ell_1$ Constrained Least Squares

### Theorem 8

Let  $K = \mathcal{B}_1$  be the unit  $\ell_1$  ball of  $\mathbb{R}^d$ ,  $d \geq 2$  and assume that  $\theta^*$ . Moreover, assume the conditions of Theorem 6 and that the columns of  $\mathbb{X}$  are normalized in such a way that  $\max_j \|\mathbb{X}_j\|_2 \leq \sqrt{n}$ . Then the constrained least squares estimator  $\hat{\theta}_{\mathcal{B}_1}^{LS}$  satisfies

$$\mathbb{E} [\text{MSE}(\mathbb{X}\hat{\theta}_{\mathcal{B}_1}^{LS})] = \frac{1}{n} \mathbb{E} \|\mathbb{X}\hat{\theta}_{\mathcal{B}_1}^{LS} - \mathbb{X}\theta^*\|_2^2 \lesssim \sigma \sqrt{\frac{\log d}{n}}.$$

Moreover, for any  $\delta > 0$ , with probability  $1 - \delta$ , it holds

$$\text{MSE}(\mathbb{X}\hat{\theta}_{\mathcal{B}_1}^{LS}) \lesssim \sigma \sqrt{\frac{\log(d/\delta)}{n}}.$$

►  $\mathcal{B}_1 = \left\{ x \in \mathbb{R}^d : \sum_{i=1}^d |x_i| \leq 1 \right\}.$

## Proof of Theorem 8.

From the considerations preceding the theorem, we got that

$$\|\mathbb{X}\hat{\theta}_{\mathcal{B}_1}^{LS} - \mathbb{X}\theta^*\|_2^2 \leq 4 \sup_{v \in \mathbb{X}K} (\epsilon^T v)^2.$$

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Note that  $K = \mathcal{B}_1 = \text{conv}(\{e_1, -e_1, \dots, e_d, -e_d\})$  and

$$\mathbb{X}K \subset \mathbf{P}_{\mathbb{X}} \triangleq \text{conv}\{\mathbb{X}_1, -\mathbb{X}_1, \dots, \mathbb{X}_d, -\mathbb{X}_d\}.$$

Therefore,

$$\|\mathbb{X}\hat{\theta}_{\mathcal{B}_1}^{LS} - \mathbb{X}\theta^*\|_2^2 \leq 4 \sup_{v \in \mathbb{X}K} (\epsilon^T v)^2 \leq 4 \sup_{v \in \mathbf{P}_{\mathbb{X}}} (\epsilon^T v)^2.$$



## Proof of Theorem 8.

From the considerations preceding the theorem, we got that

$$\|\mathbb{X}\hat{\theta}_{\mathcal{B}_1}^{LS} - \mathbb{X}\theta^*\|_2^2 \leq 4 \sup_{v \in \mathbb{X}K} (\epsilon^T v)^2.$$

Note that  $K = \mathcal{B}_1 = \text{conv}(\{e_1, -e_1, \dots, e_d, -e_d\})$  and

$$\mathbb{X}K \subset \mathbf{P}_{\mathbb{X}} \triangleq \text{conv}\{\mathbb{X}_1, -\mathbb{X}_1, \dots, \mathbb{X}_d, -\mathbb{X}_d\}.$$

Therefore,

$$\|\mathbb{X}\hat{\theta}_{\mathcal{B}_1}^{LS} - \mathbb{X}\theta^*\|_2^2 \leq 4 \sup_{v \in \mathbb{X}K} (\epsilon^T v)^2 \leq 4 \sup_{v \in \mathbf{P}_{\mathbb{X}}} (\epsilon^T v)^2.$$

Since  $\epsilon \sim \text{subG}_n(\sigma^2)$ , then for any column  $\mathbb{X}_j$  such that  $\|\mathbb{X}_j\|_2 \leq \sqrt{n}$ , the random variable  $\epsilon^T \mathbb{X}_j \sim \text{subG}(n\sigma^2)$  and so does  $-\epsilon^T \mathbb{X}_j \sim \text{subG}(n\sigma^2)$  (**Show by Yourself**). □

## Proof of Theorem 8.

Applying Theorem 5, we get the bound on  $\mathbb{E}[\text{MSE}(\mathbb{X}\hat{\theta}_K^{LS})]$ :

$$\mathbb{E}[\text{MSE}(\mathbb{X}\hat{\theta}_K^{LS})] \leq \frac{4}{n} \mathbb{E}[\sup_{v \in \mathbf{P}_{\mathbb{X}}} \epsilon^T v] \leq \frac{4}{n} \sigma \sqrt{n} \sqrt{2 \log 2d} = 4\sigma \sqrt{\frac{2 \log 2d}{n}}.$$

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And for any  $t > 0$ ,

$$\mathbb{P}(\text{MSE}(\mathbb{X}\hat{\theta}_K^{LS}) > t) \leq \mathbb{P}\left(\sup_{v \in \mathbf{P}_{\mathbb{X}}} (\epsilon^T v) > nt/4\right) \leq 2de^{-\frac{nt^2}{32\sigma^2}}$$

To conclude the proof, we find  $t$  such that

$$2de^{-\frac{nt^2}{32\sigma^2}} \leq \delta \Leftrightarrow t^2 \geq \frac{32\sigma^2}{n} \log \frac{2d}{\delta}$$

□

## $\ell_0$ constrained least squares

Define the  $\ell_0$  (pseudo) norm of vector  $\theta \in \mathbb{R}^d$  as

$$\|\theta\|_0 = \sum_{j=1}^d \mathbb{1}(\theta_j \neq 0).$$

A vector with “small”  $\ell_0$  norm is called a sparse vector. More precisely,  $\theta$  is a  $k$ -sparse vector if  $\|\theta\|_0 \leq k$ .

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The support of  $\theta$  is defined as

$$\text{supp}(\theta) = \{j \in \{1, \dots, d\} : \theta_j \neq 0\},$$

and  $\|\theta\|_0 = \text{card}(\text{supp}(\theta)) := |\text{supp}(\theta)|$ .

$$\lim_{q \rightarrow 0^+} \sum_{j=1}^d |\theta_j|^q = \|\theta\|_0.$$

## $\ell_0$ constrained least squares

Denote  $\mathcal{B}_0(k)$  the  $\ell_0$  ball of  $\mathbb{R}^d$ , i.e., the set of  $k$ -sparse, defined by

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### Theorem 9

Fix a positive integer  $k \leq d/2$ . Let  $K = \mathcal{B}_0(k)$  be set of  $k$ -sparse vectors of  $\mathbb{R}^d$  and assume that  $\theta^* \in \mathcal{B}_0(k)$ . Moreover, assume the conditions of Theorem 7. Then, for any  $\delta > 0$ , with probability  $1 - \delta$ , it holds

$$\text{MSE}(\hat{\theta}_{\mathcal{B}_0(k)}^{LS}) \lesssim \frac{\sigma^2}{n} \log \binom{d}{2k} + \frac{\sigma^2 k}{n} + \frac{\sigma^2}{n} \log(1/\delta).$$

## Proof of Theorem 9.

Similar to the proof of Theorem 7, we get

$$\|\mathbb{X}\hat{\theta}_K^{LS} - \mathbb{X}\theta^*\|_2^2 \leq 2\epsilon^T \mathbb{X}(\hat{\theta}_K^{LS} - \theta^*) = 2\|\mathbb{X}\hat{\theta}_K^{LS} - \mathbb{X}\theta^*\|_2 \frac{\epsilon^T \mathbb{X}(\hat{\theta}_K^{LS} - \theta^*)}{\|\mathbb{X}\hat{\theta}_K^{LS} - \mathbb{X}\theta^*\|_2}$$



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Since both  $\hat{\theta}_K^{LS}$  and  $\theta^*$  are in  $\mathcal{B}_0(k)$ , we have  $\hat{\theta}_K^{LS} - \theta^* \in \mathcal{B}_0(2k)$ .

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For any  $S \subset \{1, \dots, d\}$ , let  $\mathbb{X}_S$  the  $n \times |S|$  submatrix of  $\mathbb{X}$  that is obtained from the column of  $\mathbb{X}_j$ ,  $j \in S$  of  $\mathbb{X}$ . Denote by  $r_S \leq |S|$  the rank of  $\mathbb{X}_S$  and let  $\Phi_S = [\phi_1, \dots, \phi_{r_S}] \in \mathbb{R}^{n \times r_S}$  be an orthonormal basis of the column span of  $\mathbb{X}_S$ .

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Moreover, for any  $\theta \in \mathbb{R}^d$ , define  $\theta(S) \in \mathbb{R}^{|S|}$  be the vector with coordinates  $\theta_j$ ,  $j \in S$ . □

## Proof of Theorem 9.

Denote by  $\hat{S} = \text{supp}(\hat{\theta}_K^{LS} - \theta^*)$ , we have  $|\hat{S}| \leq 2k$  and there exists  $\nu \in \mathbb{R}^{r_{\hat{S}}}$  such that

$$\mathbb{X}(\hat{\theta}_K^{LS} - \theta^*) = \mathbb{X}_{\hat{S}} \left( \hat{\theta}_K^{LS}(\hat{S}) - \theta^*(\hat{S}) \right) = \Phi_{\hat{S}} \nu.$$

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Therefore,

$$\frac{\epsilon^T \mathbb{X}(\hat{\theta}_K^{LS} - \theta^*)}{\|\mathbb{X}(\hat{\theta}_K^{LS} - \theta^*)\|_2} = \frac{\epsilon^T \Phi_{\hat{S}} \nu}{\|\nu\|_2} \leq \max_{|S| \leq 2k} \sup_{u \in \mathcal{B}_2^{r_S}} [\epsilon^T \Phi_S] u = \max_{|S|=2k} \sup_{u \in \mathcal{B}_2^{r_S}} [\epsilon^T \Phi_S] u,$$

where the last equality comes from the fact the  $\mathbb{R}^{|S|}$  is the subspace of  $\mathbb{R}^{2k}$  if  $|S| \leq 2k$  and  $\mathcal{B}_2^{r_S}$  is the unit ball of  $\mathbb{R}^{r_S}$ .

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$$\|\mathbb{X} \hat{\theta}_K^{LS} - \mathbb{X} \theta^*\|_2^2 \leq 4 \max_{|S|=2k} \sup_{u \in \mathcal{B}_2^{r_S}} (\tilde{\epsilon}^T u)^2,$$

$$\tilde{\epsilon}_S = \Phi_S^T \epsilon \sim \text{subG}_{r_S}(\sigma).$$

□

## Proof of Theorem 9.

Using a union bound, we get for any  $t > 0$ ,

$$\mathbb{P} \left( \max_{|S|=2k} \sup_{u \in \mathcal{B}_2^{rS}} (\tilde{\epsilon}^T u)^2 > t \right) \leq \sum_{|S|=2k} \mathbb{P} \left( \sup_{u \in \mathcal{B}_2^{rS}} (\tilde{\epsilon}^T u)^2 > t \right)$$

It follows from the proof of Theorem 4 that for any  $|S| = 2k$ ,

$$\mathbb{P} \left( \sup_{u \in \mathcal{B}_2^{rS}} (\tilde{\epsilon}^T u)^2 > t \right) \leq 6^{2k} e^{-\frac{t}{8\sigma^2}}, \quad \forall t > 0.$$

□

## Proof of Theorem 9.

Putting together, for any  $t > 0$ ,

$$\begin{aligned}\mathbb{P}(\text{MSE}(\mathbb{X}\hat{\theta}_K^{LS}) > t) &\leq \mathbb{P}\left(\frac{4}{n} \max_{|S|=2k} \sup_{u \in \mathcal{B}_2^{r_S}} (\tilde{\epsilon}^T u)^2 > t\right) \\ &\leq \sum_{|S|=2k} \mathbb{P}\left(\sup_{u \in \mathcal{B}_2^{r_S}} (\tilde{\epsilon}^T u)^2 > nt/4\right) \\ &\leq \binom{d}{2k} 6^{2k} e^{-\frac{nt}{32\sigma^2}}\end{aligned}$$

and

$$\binom{d}{2k} 6^{2k} e^{-\frac{nt}{32\sigma^2}} \leq \delta \Leftrightarrow t > \frac{32\sigma^2}{n} \left( \log \binom{d}{2k} + 2k \log(6) + \log \frac{1}{\delta} \right)$$

□



## Lemma 10

For any integers  $1 \leq k \leq n$ , it holds

$$\binom{n}{k} \leq \left(\frac{en}{k}\right)^k.$$

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**Proof.**

Show by induction and note that  $(1 + \frac{1}{k})^k \leq e$ .



## $\ell_0$ constrained least squares

### Corollary 11

Under the assumption of Theorem 9, for any  $\delta > 0$ , with probability  $1 - \delta$ , it holds

$$\text{MSE}(\mathbb{X}\hat{\theta}_{\mathcal{B}_0(k)}^{LS}) \lesssim \frac{\sigma^2 k}{n} \log\left(\frac{ed}{2k}\right) + \frac{\sigma^2 k}{n} \log(6) + \frac{\sigma^2}{n} \log(1/\delta).$$

## $\ell_0$ constrained least squares

### Corollary 12

Under the assumption of Theorem 9,

$$\mathbb{E} \left[ \text{MSE}(\mathbb{X} \hat{\theta}_{\mathcal{B}_0(k)}^{LS}) \right] \lesssim \frac{\sigma^2 k}{n} \log \left( \frac{ed}{k} \right).$$

## Proof of Corollary 12.

For any  $H \geq 0$

$$\begin{aligned}\mathbb{E} \left[ \text{MSE}(\mathbb{X} \hat{\theta}_{\mathcal{B}_0(k)}^{LS}) \right] &= \int_0^\infty \mathbb{P}(\|\mathbb{X} \hat{\theta}_K^{LS} - \mathbb{X} \theta^*\| \geq nt) \mathrm{d}t \\ &= \int_0^H \mathbb{P}(\|\mathbb{X} \hat{\theta}_K^{LS} - \mathbb{X} \theta^*\| \geq nt) \mathrm{d}t + \int_H^\infty \mathbb{P}(\|\mathbb{X} \hat{\theta}_K^{LS} - \mathbb{X} \theta^*\| \geq nt) \mathrm{d}t\end{aligned}$$

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## Proof of Corollary 12.

For any  $H \geq 0$

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where the second inequality comes from the proof of Theorem □

## Proof of Corollary 12.

Take  $H$  such that

$$\binom{d}{2k} 6^{2k} e^{-\frac{nH}{32\sigma^2}} = 1.$$

It yields

$$H \lesssim \frac{\sigma^2 k}{n} \log \left( \frac{ed}{n} \right),$$

which completes the proof. □



# The Gaussian Sequence Model

The Gaussian sequence model is as follows:

$$Y_i = \theta_i^* + \epsilon_i, \quad i = 1, \dots, d \quad (\text{Gaussian Sequence Model})$$

where  $\epsilon_1, \dots, \epsilon_d$  are i.i.d  $\mathcal{N}(0, \sigma^2)$  random variables. The goal here is to estimate the unknown vector  $\theta^*$ .

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Note that (Gaussian Sequence Model) is a special case of the fixed design (Linear Model) with  $n = d$ , and  $\mathbb{X} = I_d$ .

# The sub-Gaussian Sequence Model

## Assumption ORT

The design matrix satisfies

$$\frac{\mathbb{X}^T \mathbb{X}}{n} = I_d,$$

where  $I_d$  denotes the identity matrix of  $\mathbb{R}^d$ .

# The sub-Gaussian Sequence Model

## Assumption ORT

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where  $I_d$  denotes the identity matrix of  $\mathbb{R}^d$ .

- ▶ Assumption ORT allows for cases where  $d \leq n$  but not  $d > n$  (high dimensional case) due to rank constraint. (**rank** $(\frac{\mathbb{X}^T \mathbb{X}}{n}) = d \leq \min\{d, n\}$ )
- ▶ The  $d$  columns of  $\mathbb{X}$  are orthogonal in  $\mathbb{R}^n$  and all have norm  $\sqrt{n}$ .

## The sub-Gaussian Sequence Model

Under Assumption ORT, it follows from (Linear Model) that

$$\begin{aligned}y &:= \frac{1}{n} \mathbb{X}^T Y = \frac{\mathbb{X}^T \mathbb{X}}{n} \theta^* + \frac{1}{n} \mathbb{X}^T \epsilon \\&= \theta^* + \xi,\end{aligned}$$

where  $\xi = (\xi_1, \dots, \xi_d) \sim \text{subG}_d(\sigma^2/n)$ .

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where  $\xi = (\xi_1, \dots, \xi_d) \sim \text{subG}_d(\sigma^2/n)$ .

Under Assumption ORT,

- ▶ The Linear Model is equivalent to the sub-Gaussian Sequence Model Gaussian Sequence Model up a transformation of the data  $Y$  and a change of variable for the variance.
- ▶ For any  $\hat{\theta} \in \mathbb{R}^d$ ,

$$\text{MSE}(\mathbb{X}\hat{\theta}) = (\hat{\theta} - \theta^*)^T \frac{\mathbb{X}^T \mathbb{X}}{n} (\hat{\theta} - \theta^*) = \|\hat{\theta} - \theta^*\|_2^2$$

## The sub-Gaussian Sequence Model

For any  $\theta \in \mathbb{R}^d$ , Assumption ORT yields,

$$\begin{aligned}\|y - \theta\|_2^2 &= \left\| \frac{1}{n} \mathbb{X}^T Y - \theta \right\|_2^2 \\ &= \|\theta\|_2^2 - \frac{2}{n} \theta^T \mathbb{X} Y + \frac{1}{n^2} Y^T \mathbb{X} \mathbb{X}^T Y \\ &= \frac{1}{n} \|\mathbb{X} \theta\|_2^2 - \frac{2}{n} (\mathbb{X} \theta)^T Y + \frac{1}{n} \|Y\|_2^2 + Q \\ &= \frac{1}{n} \|Y - \mathbb{X} \theta\|_2^2 + Q,\end{aligned}$$

(Equivalence to LSE)

where  $Q$  is a constant that does not depend on  $\theta$  and is defined by

$$Q = \frac{1}{n^2} = \frac{1}{n^2} Y^T \mathbb{X} \mathbb{X}^T Y - \frac{1}{n} \|Y\|_2^2.$$

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This implies that the least squares estimator  $\hat{\theta}^{LS}$  is equal to  $y$  (By the optimal condition).



# The sub-Gaussian Sequence Model

The preceding discussion to be summarized by a slightly more general model called sub – Gaussian sequence model:

$$y = \theta^* + \xi \quad \in \mathbb{R}^d \quad \text{(Sub-Gaussian Sequence Model)}$$

where  $\xi \sim \text{subG}_d(\sigma^2/n)$ .

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where  $\xi \sim \text{subG}_d(\sigma^2/n)$ .

- We define this model independently of Assumption ORT and thus for any values of  $n$  and  $d$ .

# Sparsity Adaptive Thresholding Estimators

If we knew a prior that  $\theta$  was  $k$  sparse, we could employ directly Corollary 11 to obtain that with probability  $1 - \delta$ , we have

$$\text{MSE}(\hat{\theta}_{\mathcal{B}_0(k)}^{LS}) \leq C_\delta \frac{\sigma^2 k}{n} \log \left( \frac{ed}{2k} \right). \quad (\ell_0 \text{ constraint estimator with known sparsity})$$

# Sparsity Adaptive Thresholding Estimators

Assume the (Sub-Gaussian Sequence Model). If nothing is known about  $\theta^*$ , it is natural to estimate it using the least squares estimator  $\hat{\theta}^{LS} = y$ . In this case,

$$\text{MSE}(\hat{\theta}^{LS}) = \|y - \theta^*\|_2^2 = \|\xi\| \leq C_\delta \frac{\sigma^2 d}{n},$$

where the last inequality holds with probability at least  $1 - \delta$ . It is consistent with ( $\ell_0$  constraint estimator with known sparsity) if  $k = Cd$  for some positive constant  $C \leq 1$ .

# Sparsity Adaptive Thresholding Estimators

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where the last inequality holds with probability at least  $1 - \delta$ . It is consistent with ( $\ell_0$  constraint estimator with known sparsity) if  $k = Cd$  for some positive constant  $C \leq 1$ .

However, this approach does not use the fact that  $k$  may be much smaller than  $d$ , which happens when  $\theta^*$  has many zero coordinate.

## Sparsity Adaptive Thresholding Estimators

If  $\theta_j^* = 0$ , then,  $y_j = \xi_j$ , a sub-Gaussian variable with variance proxy  $\sigma^2/n$ . In particular, we know that with probability at least  $1 - \delta$ ,

$$|\xi_j| \leq \sigma \sqrt{\frac{2 \log(2/\delta)}{n}} = \tau.$$

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The consequences of this inequality are interesting:

► **If we observe  $|y_i| \gg \tau$ , then it must correspond to  $\theta_j^* \neq 0$ .**

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The consequences of this inequality are interesting:

- ▶ **If we observe  $|y_i| \gg \tau$ , then it must correspond to  $\theta_j^* \neq 0$ .**
- ▶ **If  $|y_j| \leq \tau$  is smaller, then  $\theta_j^*$  cannot be very large. By the triangle inequality,  $|\theta_j^*| \leq |y_j| + |\xi_j| \leq 2\tau$ . Therefore, we loose at most  $2\tau$  by choosing  $\hat{\theta}_j = 0$ .**



## Sparsity Adaptive Thresholding Estimators

If  $\theta_j^* = 0$ , then,  $y_j = \xi_j$ , a sub-Gaussian variable with variance proxy  $\sigma^2/n$ . In particular, we know that with probability at least  $1 - \delta$ ,

$$|\xi_j| \leq \sigma \sqrt{\frac{2 \log(2/\delta)}{n}} = \tau.$$

The consequences of this inequality are interesting:

- ▶ **If we observe  $|y_i| \gg \tau$ , then it must correspond to  $\theta_j^* \neq 0$ .**
- ▶ **If  $|y_j| \leq \tau$  is smaller, then  $\theta_j^*$  cannot be very large. By the triangle inequality,  $|\theta_j^*| \leq |y_j| + |\xi_j| \leq 2\tau$ . Therefore, we loose at most  $2\tau$  by choosing  $\hat{\theta}_j = 0$ .**

Such observations lead us to consider the **hard thresholding estimator**.

# Sparsity Adaptive Thresholding Estimators

## Definition 13 (Hard Thresholding Estimator)

The hard thresholding estimator with threshold  $2\tau > 0$  is denoted by  $\hat{\theta}^{\text{HRD}}$  and has coordinates

$$\hat{\theta}_j^{\text{HRD}} = \begin{cases} y_j & \text{if } |y_j| > 2\tau, \\ 0 & \text{if } |y_j| \leq 2\tau, \end{cases}$$

for  $j = 1, \dots, d$ . In short, we can write  $\hat{\theta}_j^{\text{HRD}} = y_j \mathbb{1}(|y_j| > 2\tau)$

# Sparsity Adaptive Thresholding Estimators

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for  $j = 1, \dots, d$ . In short, we can write  $\hat{\theta}_j^{\text{HRD}} = y_j \mathbb{1}(|y_j| > 2\tau)$

For each individual  $\xi_j$ , the inequality

$$|\xi_j| \leq \sigma \sqrt{\frac{2 \log(2/\delta)}{n}} \quad \text{holds with prob. at least } 1 - \delta.$$

For the hard thresholding estimator, we should consider  $\tau$  such that  $|\xi_j| \leq \tau$  holds simultaneously for all  $j$ , which can be achieved by controlling the maxima.

# Sparsity Adaptive Thresholding Estimators

By the inequality of suprema of sub-Gaussian variables, we have

$$\max_{1 \leq j \leq d} |\xi_j| \leq \sigma \sqrt{\frac{2 \log(2d/\delta)}{n}} \quad \text{holds with prob. at least } 1 - \delta.$$

It yields the following theorem.

# Sparsity Adaptive Thresholding Estimators

## Theorem 14

Consider the (Linear Model) under the Assumption (ORT) or, equivalently, the (Sub-Gaussian Sequence Model). Then the hard thresholding estimator  $\hat{\theta}^{\text{HRD}}$  with threshold

$$2\tau = 2\sigma\sqrt{\frac{2\log(2d/\delta)}{n}}$$

enjoys the following two properties on the same event  $\mathcal{A}$  such that  $\mathbb{P}(\mathcal{A}) \geq 1 - \delta$ :

(1) If  $\|\theta\|_0 = k$ ,

$$\text{MSE}(\mathbb{X}\hat{\theta}^{\text{HRD}}) = \|\hat{\theta}^{\text{HRD}} - \theta^*\|_2^2 \lesssim \sigma^2 \frac{k \log(2d/\delta)}{n}.$$

(2) If  $\min_{j \in \text{supp}(\theta^*)} > 3\tau$ , then

$$\text{supp}(\hat{\theta}^{\text{HRD}}) = \text{supp}(\theta^*).$$

## Proof of Theorem 14.

Define the event

$$\mathcal{A} = \left\{ \max_j |\xi_j| \leq \tau \right\}$$

and by the inequality of maxima of sub-Gaussian variables  $\mathbb{P}(\mathcal{A}) \geq 1 - \delta$ .

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Define the event

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and by the inequality of maxima of sub-Gaussian variables  $\mathbb{P}(\mathcal{A}) \geq 1 - \delta$ . On the event  $\mathcal{A}$ , the following holds for any  $j = 1, \dots, d$ . First, observe that

$$|y_j| > 2\tau \Rightarrow |\theta_j^*| \geq |y_j| - |\xi_j| > \tau \quad (\clubsuit)$$

and

$$|y_j| \leq 2\tau \Rightarrow |\theta_j^*| \leq |y_j| + |\xi_j| \leq 3\tau. \quad (\spadesuit)$$



## Proof of Theorem 14.

It yields

$$\begin{aligned} |\hat{\theta}_j^{\text{HRD}} - \theta_j^*| &= |y_j - \theta_j^*| \mathbb{1}(|y_j| > 2\tau) + |\theta_j^*| \mathbb{1}(|y_j| \leq 2\tau) \\ &\leq \tau \mathbb{1}(|y_j| > 2\tau) + |\theta_j^*| \mathbb{1}(|y_j| \leq 2\tau) \end{aligned}$$



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$$\begin{aligned} |\hat{\theta}_j^{\text{HRD}} - \theta_j^*| &= |y_j - \theta_j^*| \mathbb{1}(|y_j| > 2\tau) + |\theta_j^*| \mathbb{1}(|y_j| \leq 2\tau) \\ &\leq \tau \mathbb{1}(|y_j| > 2\tau) + |\theta_j^*| \mathbb{1}(|y_j| \leq 2\tau) \\ &\leq \tau \mathbb{1}(|\theta_j^*| > \tau) + |\theta_j^*| \mathbb{1}(|\theta_j^*| \leq 3\tau) \\ &\leq 4 \min(|\theta_j^*|, \tau). \end{aligned}$$

by (♣) and (♠)

**(Check by Yourself)**

## Proof of Theorem 14.

It yields

$$\begin{aligned} |\hat{\theta}_j^{\text{HRD}} - \theta_j^*| &= |y_j - \theta_j^*| \mathbb{1}(|y_j| > 2\tau) + |\theta_j^*| \mathbb{1}(|y_j| \leq 2\tau) \\ &\leq \tau \mathbb{1}(|y_j| > 2\tau) + |\theta_j^*| \mathbb{1}(|y_j| \leq 2\tau) \\ &\leq \tau \mathbb{1}(|\theta_j^*| > \tau) + |\theta_j^*| \mathbb{1}(|\theta_j^*| \leq 3\tau) \\ &\leq 4 \min(|\theta_j^*|, \tau). \end{aligned}$$

by (♣) and (♠)

**(Check by Yourself)**

It yields

$$\begin{aligned} \|\hat{\theta}^{\text{HRD}} - \theta^*\|_2^2 &= \sum_{j=1}^d 6d |\hat{\theta}_j^{\text{HRD}} - \theta_j^*|^2 \\ &\leq 16 \sum_{j=1}^d \min(|\theta_j^*|^2, \tau^2) \leq 16 \|\theta\|_0 \tau^2. \end{aligned}$$

This complete the proof of (1).



## Proof of Theorem 14.

To prove (2), note that if  $\theta_j^* \neq 0$ , then  $|\theta_j^*| > 3\tau$ . It yields

$$|y_j| = |\theta_j^* + \xi| \geq |\theta_j^*| - |\xi| > 3\tau - \tau = 2\tau.$$

Therefore  $\hat{\theta}_j^{\text{HRD}} \neq 0$  so that  $\text{supp}(\theta^*) \subset \text{supp}(\hat{\theta}^{\text{HRD}})$ .

## Proof of Theorem 14.

To prove (2), note that if  $\theta_j^* \neq 0$ , then  $|\theta_j^*| > 3\tau$ . It yields

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Therefore  $\hat{\theta}_j^{\text{HRD}} \neq 0$  so that  $\text{supp}(\theta^*) \subset \text{supp}(\hat{\theta}^{\text{HRD}})$ .

Next, if  $\hat{\theta}_j^{\text{HRD}} \neq 0$ , then  $|\hat{\theta}_j^{\text{HRD}}| = |y_j| > 2\tau$ . It yields

$$|\theta_j^*| = |y_j - \xi_j| > 2\tau - \tau > \tau.$$

Therefore,  $|\theta_j^*| \neq 0$  and  $\text{supp}(\hat{\theta}^{\text{HRD}}) \subset \text{supp}(\theta^*)$ .

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To prove (2), note that if  $\theta_j^* \neq 0$ , then  $|\theta_j^*| > 3\tau$ . It yields

$$|y_j| = |\theta_j^* + \xi| \geq |\theta_j^*| - |\xi| > 3\tau - \tau = 2\tau.$$

Therefore  $\hat{\theta}_j^{\text{HRD}} \neq 0$  so that  $\text{supp}(\theta^*) \subset \text{supp}(\hat{\theta}^{\text{HRD}})$ .

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Therefore,  $|\theta_j^*| \neq 0$  and  $\text{supp}(\hat{\theta}^{\text{HRD}}) \subset \text{supp}(\theta^*)$ .

Hence, it concludes that  $\text{supp}(\hat{\theta}^{\text{HRD}}) = \text{supp}(\theta^*)$

□

# Sparsity Adaptive Thresholding Estimators

Similar results can be obtained for the **soft thresholding** estimator  $\hat{\theta}^{\text{SFT}}$  defined by

$$\hat{\theta}_j^{\text{SFT}} = \begin{cases} y_j - 2\tau & \text{if } y_j > 2\tau, \\ y_j + 2\tau & \text{if } y_j < -2\tau, \\ 0 & \text{if } |y_j| \leq 2\tau, \end{cases}$$

In short, we can write

$$\hat{\theta}_j^{\text{SFT}} = \left(1 - \frac{2\tau}{|y_j|}\right)_+ y_j.$$

# The BIC and Lasso Estimators

It can be shown that the hard and soft thresholding estimators are solutions of the following penalized empirical risk minimization problems:

$$\hat{\theta}^{\text{HRD}} = \arg \min_{\theta \in \mathbb{R}^d} \{ \|y - \theta\|_2^2 + 4\tau^2 \|\theta\|_0 \}$$

$$\hat{\theta}^{\text{SFT}} = \arg \min_{\theta \in \mathbb{R}^d} \{ \|y - \theta\|_2^2 + 4\tau \|\theta\|_1 \}$$

# The BIC and Lasso Estimators

In view of (Equivalence to LSE), under the Assumption ORT, the variational definitions can be written as

$$\hat{\theta}^{\text{HRD}} = \arg \min \left\{ \frac{1}{n} \|Y - X\theta\|_2^2 + 4\tau^2 \|\theta\|_0 \right\}$$

$$\hat{\theta}^{\text{SFT}} = \arg \min \left\{ \frac{1}{n} \|Y - X\theta\|_2^2 + 4\tau \|\theta\|_1 \right\}$$



# The BIC and Lasso Estimators

## Definition 15

Fix  $\tau > 0$  and assume (Linear Model). The BIC(Bayes Information Criterion) estimator of  $\theta^*$  in is defined by an  $\hat{\theta}^{\text{BIC}}$  such that

$$\hat{\theta}^{\text{BIC}} \in \arg \min_{\theta \in \mathbb{R}^d} \left\{ \frac{1}{n} \|Y - \mathbb{X}\theta\|_2^2 + \tau^2 \|\theta\|_0 \right\}.$$

Moreover the Lasso estimator of  $\theta^*$  in is defined by any  $\hat{\theta}^{\mathcal{L}}$  such that

$$\hat{\theta}^{\mathcal{L}} \in \arg \min_{\theta \in \mathbb{R}^d} \left\{ \frac{1}{n} \|Y - \mathbb{X}\theta\|_2^2 + 2\tau \|\theta\|_1 \right\}.$$

# The BIC and Lasso Estimators

Computing the BIC estimator can be proved to be NP-hard in the worst case. In particular, no computational method is known to be significantly faster than the brute force search among all  $2^d$  sparsity patterns.

$$\min_{\theta \in \mathbb{R}^d} \left\{ \frac{1}{n} \|Y - \mathbb{X}\theta\|_2^2 + \tau^2 \|\theta\|_0 \right\} = \min_{0 \leq k \leq d} \left\{ \min_{\theta: \|\theta\|_0 = k} \frac{1}{n} \|Y - \mathbb{X}\theta\|_2^2 + \tau^2 \|\theta\|_0 \right\}$$

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To compute  $\min_{\theta: \|\theta\|_0 = k} \frac{1}{n} \|Y - \mathbb{X}\theta\|_2^2$ , we need to compute  $\binom{d}{k}$  least squares estimators on a space of size  $k$ . Each costs  $O(k^3)$  (matrix inversion).

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To compute  $\min_{\theta: \|\theta\|_0 = k} \frac{1}{n} \|Y - \mathbb{X}\theta\|_2^2$ , we need to compute  $\binom{d}{k}$  least squares estimators on a space of size  $k$ . Each costs  $O(k^3)$  (matrix inversion). Therefore, the total cost of the brute force search is

$$C \sum_{k=0}^d \binom{d}{k} k^3 = \Theta(d^3 2^d).$$

# The BIC and Lasso Estimators

Solvers for the Lasso estimator:

- ▶ Coordinate gradient descent implemented in the **glmnet** in **R**.
- ▶ LARS that computes the entire regularization path, i.e., the solution of the convex problem for all value of  $\tau$ . It relies on the fact that, as a function of  $\tau$ , the solution  $\hat{\theta}^{\mathcal{L}}$  is a piecewise linear function (with values in  $\mathbb{R}^d$ .) LARS is slow for very large problem.
- ▶ FISTA.
- ▶ Stochastic gradient descent for very large  $d$  and very large  $n$  when computing  $\|Y - \mathbb{X}\theta\|_2^2$  may be computationally expensive.

# Analysis of the BIC estimator

## Theorem 16

Assume that the (Linear Model) holds such that  $\epsilon \sim \text{subG}_n(\sigma^2)$ . Then, the BIC estimator  $\hat{\theta}^{\text{BIC}}$  with regularization parameter

$$\tau^2 = 16 \log(6) \frac{\sigma^2}{n} + 32 \frac{\sigma^2 \log(ed)}{n} \quad (\spadesuit)$$

satisfies

$$\text{MSE}(\mathbb{X} \hat{\theta}^{\text{BIC}}) = \frac{1}{n} \|\mathbb{X} \hat{\theta}^{\text{BIC}} - \mathbb{X} \theta^*\|_2^2 \lesssim \|\theta^*\|_0 \sigma^2 \frac{\log(ed/\delta)}{n}$$

with probability at least  $1 - \delta$ .

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with probability at least  $1 - \delta$ .

- The theorem shows that  $\hat{\theta}^{\text{BIC}}$  adapts to the unknown sparsity of  $\theta^*$ , just like  $\hat{\theta}^{\text{HRD}}$ . And it holds under no assumption on the design matrix  $\mathbb{X}$ .

## Proof of Theorem 16.

We begin as usual by noting that

$$\frac{1}{n}\|Y - \mathbb{X}\hat{\theta}^{\text{BIC}}\|_2^2 + \tau^2\|\hat{\theta}^{\text{BIC}}\|_0 \leq \frac{1}{n}\|Y - \mathbb{X}\theta^*\|_2^2 + \tau^2\|\theta^*\|_0 \leq \frac{1}{n}\|\epsilon\|_2^2 + \tau^2\|\theta^*\|_0$$



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It implies

$$\|\mathbb{X}\hat{\theta}^{\text{BIC}} - \mathbb{X}\theta^*\|_2^2 \leq n\tau^2\|\theta^*\|_0 + 2\epsilon^T\mathbb{X}(\hat{\theta}^{\text{BIC}} - \theta^*) - n\tau^2\|\hat{\theta}^{\text{BIC}}\|_0$$



## Proof of Theorem 16.

First, note that

$$\begin{aligned} 2\epsilon^T(\mathbb{X}\hat{\theta}^{\text{BIC}} - \mathbb{X}\theta^*) &= 2\epsilon^T \left( \frac{\mathbb{X}\hat{\theta}^{\text{BIC}} - \mathbb{X}\theta^*}{\|\mathbb{X}\hat{\theta}^{\text{BIC}} - \mathbb{X}\theta^*\|_2^2} \right) \|\mathbb{X}\hat{\theta}^{\text{BIC}} - \mathbb{X}\theta^*\|_2 \\ &\leq 2 \left[ \epsilon^T \left( \frac{\mathbb{X}\hat{\theta}^{\text{BIC}} - \mathbb{X}\theta^*}{\|\mathbb{X}\hat{\theta}^{\text{BIC}} - \mathbb{X}\theta^*\|_2^2} \right) \right]^2 + \frac{1}{2} \|\mathbb{X}\hat{\theta}^{\text{BIC}} - \mathbb{X}\theta^*\|_2^2 \end{aligned}$$

where we used the inequality  $2ab = 2(\sqrt{2a} \cdot \frac{1}{\sqrt{2}}b) \leq 2a^2 + \frac{1}{2}b^2$ .

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where we used the inequality  $2ab = 2(\sqrt{2a} \cdot \frac{1}{\sqrt{2}}b) \leq 2a^2 + \frac{1}{2}b^2$ .

Putting together, it yields

$$\|\mathbb{X}\hat{\theta}^{\text{BIC}} - \mathbb{X}\theta^*\|_2^2 \leq 2n\tau^2 \|\theta^*\|_0 + 4 \left[ \epsilon^T \mathcal{U}(\hat{\theta}^{\text{BIC}} - \theta^*) \right]^2 - 2n\tau^2 \|\hat{\theta}^{\text{BIC}}\|_0 \quad (\clubsuit)$$

where  $\mathcal{U}(\hat{\theta}^{\text{BIC}} - \theta^*) = \frac{\mathbb{X}\hat{\theta}^{\text{BIC}} - \mathbb{X}\theta^*}{\|\mathbb{X}\hat{\theta}^{\text{BIC}} - \mathbb{X}\theta^*\|_2^2}$ . □

## Proof of Theorem 16.

Next, we need to “sup out”  $\hat{\theta}^{\text{BIC}}$ . To that end, we decompose the sup into a max over cardinalities as follows:

$$\sup_{\theta \in \mathbb{R}^d} = \max_{1 \leq k \leq d} \max_{|S|=k} \sup_{\text{supp}(\theta)=S} .$$

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Applied to the above inequality, it yields

$$\begin{aligned} & 4 \left[ \epsilon^T \mathcal{U}(\hat{\theta}^{\text{BIC}} - \theta^*) \right]^2 - 2n\tau^2 \|\hat{\theta}^{\text{BIC}}\|_0 \\ & \leq \max_{1 \leq k \leq d} \left\{ \max_{|S|=k} \sup_{\text{supp}(\theta)=S} 4 \left[ \epsilon^T \mathcal{U}(\theta - \theta^*) \right]^2 - 2n\tau^2 k \right\} \end{aligned}$$

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$$\begin{aligned} & 4 \left[ \epsilon^T \mathcal{U}(\hat{\theta}^{\text{BIC}} - \theta^*) \right]^2 - 2n\tau^2 \|\hat{\theta}^{\text{BIC}}\|_0 \\ & \leq \max_{1 \leq k \leq d} \left\{ \max_{|S|=k} \sup_{\text{supp}(\theta)=S} 4 \left[ \epsilon^T \mathcal{U}(\theta - \theta^*) \right]^2 - 2n\tau^2 k \right\} \\ & \leq \max_{1 \leq k \leq d} \left\{ \max_{|S|=k} \sup_{u \in \mathcal{B}_2^{r_{S,*}}} 4 \left[ \epsilon^T \Phi_{S,*} u \right]^2 - 2n\tau^2 k \right\}, \end{aligned}$$

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Applied to the above inequality, it yields

$$\begin{aligned} & 4 \left[ \epsilon^T \mathcal{U}(\hat{\theta}^{\text{BIC}} - \theta^*) \right]^2 - 2n\tau^2 \|\hat{\theta}^{\text{BIC}}\|_0 \\ & \leq \max_{1 \leq k \leq d} \left\{ \max_{|S|=k} \sup_{\text{supp}(\theta)=S} 4 \left[ \epsilon^T \mathcal{U}(\theta - \theta^*) \right]^2 - 2n\tau^2 k \right\} \\ & \leq \max_{1 \leq k \leq d} \left\{ \max_{|S|=k} \sup_{u \in \mathcal{B}_2^{r_{S,*}}} 4 \left[ \epsilon^T \Phi_{S,*} u \right]^2 - 2n\tau^2 k \right\}, \end{aligned}$$

where  $\Phi_{S,*} = [\phi_1, \dots, \phi_{r_{S,*}}]$  is an orthonormal basis of the set  $\{\mathbb{X}_j, j \in S \cup \text{supp}(\theta^*)\}$  of columns of  $\mathbb{X}$  and  $r_{S,*} \leq |S| + \|\theta^*\|_0$  is the dimension of this column span.  $\square$

## Proof of Theorem 16.

Using the union bounds, we get for any  $t > 0$ ,

$$\begin{aligned} & \mathbb{P} \left( \max_{1 \leq k \leq d} \left\{ \max_{|S|=k} \sup_{u \in \mathcal{B}_2^{r_{S,*}}} 4 [\epsilon^T \Phi_{S,*} u]^2 - 2n\tau^2 k \right\} \geq t \right) \\ & \leq \sum_{k=1}^d \sum_{|S|=k} \mathbb{P} \left( \sup_{u \in \mathcal{B}_2^{r_{S,*}}} [\epsilon^T \Phi_{S,*} u]^2 \geq \frac{t}{4} + \frac{1}{2} n\tau^2 k \right) \end{aligned}$$

□



## Proof of Theorem 16.

Moreover, from Theorem 4, we get for  $|S| = k$ ,

$$\mathbb{P} \left( \sup_{u \in \mathcal{B}_2^{r_{S,*}}} [\epsilon^T \Phi_{S,*} u]^2 \geq \frac{t}{4} + \frac{1}{2} n \tau^2 k \right) \leq 2 \cdot 6^{r_{S,*}} \exp \left( -\frac{t/4 + n \tau^2 k/2}{8 \sigma^2} \right)$$

## Proof of Theorem 16.

Moreover, from Theorem 4, we get for  $|S| = k$ ,

$$\begin{aligned} \mathbb{P} \left( \sup_{u \in \mathcal{B}_2^{r_{S,*}}} [\epsilon^T \Phi_{S,*} u]^2 \geq \frac{t}{4} + \frac{1}{2} n \tau^2 k \right) &\leq 2 \cdot 6^{r_{S,*}} \exp \left( -\frac{t/4 + n \tau^2 k/2}{8 \sigma^2} \right) \\ &\leq 2 \exp \left( -\frac{t}{32 \sigma^2} - \frac{n \tau^2 k}{16 \sigma^2} + (k + \|\theta^*\|_0) \log(6) \right) \quad (r_{S,*} \leq |S| + \|\theta^*\|_0) \\ &\leq \exp \left( -\frac{t}{32 \sigma^2} - 2k \log(ed) + \|\theta^*\|_0 \log(12) \right) \end{aligned}$$

where, in the last inequality, we used the definition ( $\spadesuit$ ) of  $\tau$ . □

## Proof of Theorem 16.

Putting together, we get

$$\begin{aligned}\mathbb{P}\left(\|\mathbb{X}\hat{\theta}^{\text{BIC}} - \mathbb{X}\theta^*\|_2^2 \geq 2n\tau^2\|\theta^*\|_0 + t\right) &\leq \sum_{k=1}^d \sum_{|S|=k} \exp\left(-\frac{t}{32\sigma^2} - 2k\log(ed) + \|\theta^*\|_0 \log(12)\right) \\ &= \sum_{k=1}^d \binom{d}{k} \exp\left(-\frac{t}{32\sigma^2} - 2k\log(ed) + \|\theta^*\|_0 \log(12)\right)\end{aligned}$$

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## Proof of Theorem 16.

Putting together, we get

$$\begin{aligned}\mathbb{P}\left(\|\mathbb{X}\hat{\theta}^{\text{BIC}} - \mathbb{X}\theta^*\|_2^2 \geq 2n\tau^2\|\theta^*\|_0 + t\right) &\leq \sum_{k=1}^d \sum_{|S|=k} \exp\left(-\frac{t}{32\sigma^2} - 2k\log(ed) + \|\theta^*\|_0 \log(12)\right) \\ &= \sum_{k=1}^d \binom{d}{k} \exp\left(-\frac{t}{32\sigma^2} - 2k\log(ed) + \|\theta^*\|_0 \log(12)\right) \\ &\leq \sum_{k=1}^d \exp\left(-\frac{t}{32\sigma^2} - k\log(ed) + \|\theta^*\|_0 \log(12)\right) \\ &= \sum_{k=1}^d (ed)^{-k} \exp\left(-\frac{t}{32\sigma^2} + \|\theta^*\|_0 \log(12)\right) \\ &\leq \exp\left(-\frac{t}{32\sigma^2} + \|\theta^*\|_0 \log(12)\right)\end{aligned}$$

## Proof of Theorem 16.

Putting together, we get

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where, in the third inequality, we apply  $\binom{d}{k} \leq \left(\frac{ed}{k}\right)^k$ .

□

## Proof of Theorem 16.

To conclude the proof, choose  $t = 32\sigma^2\|\theta^*\|_0 \log(12) + 32\sigma^2 \log(1/\delta)$  and observe that combined with (♣), it yields with probability  $1 - \delta$

$$\begin{aligned}\|\mathbb{X}\hat{\theta}^{\text{BIC}} - \mathbb{X}\theta^*\|_2^2 &\leq 2n\tau^2\|\theta\|_0^* + t \\ &= 64\sigma^2 \log(ed)\|\theta^*\|_0 + 64\log(12)\sigma^2\|\theta^*\|_0 + 32\sigma^2 \log(1/\delta) \\ &\leq 224\|\theta^*\|_0\sigma^2 \log(ed) + 32\sigma^2 \log(1/\delta)\end{aligned}$$

where we have used the definition (♠) of  $\tau$ . □

# Slow rate for the Lasso estimator

## Theorem 17

Assume that the (Linear Model) holds where  $\epsilon \sim \text{subG}(\sigma^2)$ . Moreover, assume that the columns of  $\mathbb{X}$  are normalized in such a way that  $\max_j \|\mathbb{X}_j\|_2 \leq \sqrt{n}$ . Then, the Lasso estimator  $\hat{\theta}^{\mathcal{L}}$  with regularization parameter

$$2\tau = 2\sigma\sqrt{\frac{2\log(ed)}{n}} + 2\sigma\sqrt{\frac{2\log(1/\delta)}{n}} \quad (\diamond)$$

satisfies

$$\text{MSE}(\mathbb{X}\hat{\theta}^{\mathcal{L}}) = \frac{1}{n}\|\mathbb{X}\hat{\theta}^{\mathcal{L}} - \mathbb{X}\theta^*\|_2^2 \leq 4\|\theta^*\|_1\sigma\sqrt{\frac{2\log(2d)}{n}} + 4\|\theta^*\|_1\sigma\sqrt{\frac{2\log(1/\delta)}{n}}$$

with probability  $1 - \delta$ . Moreover, there exists a numerical constant  $C > 0$  such that

$$\mathbb{E}[\text{MSE}(\mathbb{X}\hat{\theta}^{\mathcal{L}})] \leq C\|\theta^*\|_1\sigma\sqrt{\frac{\log(2d)}{n}}.$$



## Proof of Theorem 17.

From the definition of  $\hat{\theta}^{\mathcal{L}}$ , it holds

$$\frac{1}{n}\|Y - \mathbb{X}\hat{\theta}^{\mathcal{L}}\|_2^2 + 2\tau\|\hat{\theta}^{\mathcal{L}}\|_1 \leq \frac{1}{n}\|Y - \mathbb{X}\theta^*\|_2^2 + 2\tau\|\theta^*\|_1.$$

## Proof of Theorem 17.

From the definition of  $\hat{\theta}^{\mathcal{L}}$ , it holds

$$\frac{1}{n}\|Y - \mathbb{X}\hat{\theta}^{\mathcal{L}}\|_2^2 + 2\tau\|\hat{\theta}^{\mathcal{L}}\|_1 \leq \frac{1}{n}\|Y - \mathbb{X}\theta^*\|_2^2 + 2\tau\|\theta^*\|_1.$$

It implies

$$\begin{aligned}\|\mathbb{X}\hat{\theta}^{\mathcal{L}} - \mathbb{X}\theta^*\|_2^2 &\leq 2\epsilon^T \mathbb{X}(\hat{\theta}^{\mathcal{L}} - \theta^*) + 2n\tau(\|\theta^*\|_1 - \|\hat{\theta}^{\mathcal{L}}\|_1) \\ &\leq 2\|\mathbb{X}^T \epsilon\|_{\infty} \|\hat{\theta}^{\mathcal{L}}\|_1 - 2n\tau\|\hat{\theta}^{\mathcal{L}}\|_1 + 2\|\mathbb{X}^T \epsilon\|_{\infty} \|\theta^*\|_1 + 2n\tau\|\theta^*\|_1 \\ &= 2(\|\mathbb{X}^T \epsilon\|_{\infty} - n\tau)\|\hat{\theta}^{\mathcal{L}}\|_1 + 2(\|\mathbb{X}^T \epsilon\|_{\infty} + n\tau)\|\theta^*\|_1,\end{aligned}$$

where the Hölder's inequality is used. □

## Proof of Theorem 17.

Since  $\max_j \|\mathbb{X}_j\|_2 \leq \sqrt{n}$ , we have  $\mathbb{X}_j^T \epsilon \sim \text{subG}(n\sigma^2)$ .

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$$\mathbb{P}(\|\mathbb{X}^T \epsilon\|_\infty > t) = \mathbb{P}(\max_{1 \leq j \leq d} |\mathbb{X}_j^T \epsilon| > t)$$

## Proof of Theorem 17.

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$$\begin{aligned}\mathbb{P}(\|\mathbb{X}^T \epsilon\|_\infty > t) &= \mathbb{P}(\max_{1 \leq j \leq d} |\mathbb{X}_j^T \epsilon| > t) \\ &\leq \sum_{j=1}^d \mathbb{P}(|\mathbb{X}_j^T \epsilon| > t) \quad (\text{Union Bound}) \\ &\leq 2d \exp\left(-\frac{t^2}{2n\sigma^2}\right)\end{aligned}$$

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Therefore, taking  $t = \sigma\sqrt{2n \log(2d)} + \sigma\sqrt{2n \log(1/\delta)} = n\tau$ , we get that with probability  $1 - \delta$ ,

$$\|\mathbb{X}\hat{\theta}^{\mathcal{L}} - \mathbb{X}\theta^*\|_2^2 \leq 4n\tau\|\theta^*\|_1.$$

□

## Proof of Theorem 17.

Let

$$Z = \text{MSE}(\mathbb{X}\hat{\theta}^{\mathcal{L}}) = \frac{1}{n} \|\mathbb{X}\hat{\theta}^{\mathcal{L}} - \mathbb{X}\theta^*\|_2^2.$$

## Proof of Theorem 17.

Let

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From the high-probability bound, for any  $\xi > 0$ ,

$$\mathbb{P} \left( Z \geq \|\theta^*\|_1 \sigma \sqrt{\frac{2 \log(2d)}{n}} + 4 \|\theta^*\|_1 \sigma \sqrt{\frac{2\xi}{n}} \right) \leq e^{-\xi}.$$



## Proof of Theorem 17.

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$$\mathbb{P} \left( Z \geq \|\theta^*\|_1 \sigma \sqrt{\frac{2 \log(2d)}{n}} + 4\|\theta^*\|_1 \sigma \sqrt{\frac{2\xi}{n}} \right) \leq e^{-\xi}.$$

Define

$$A = \|\theta^*\|_1 \sigma \sqrt{\frac{2 \log(2d)}{n}}, \quad B = 4\|\theta^*\|_1 \sigma \sqrt{\frac{2}{n}}.$$

Then

$$\mathbb{P}(Z \geq A + B\sqrt{\xi}) \leq e^{-\xi}.$$

□

## Proof of Theorem 17.

Equivalently, for any  $t \geq A$ ,

$$\mathbb{P}(Z \geq t) \leq \exp \left\{ - \left( \frac{t - A}{B} \right)^2 \right\}.$$

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$$\mathbb{P}(Z \geq t) \leq \exp \left\{ - \left( \frac{t - A}{B} \right)^2 \right\}.$$

Hence, we obtain

$$\begin{aligned} \mathbb{E}[Z] &= \int_0^A \mathbb{P}(Z \geq t) dt + \int_A^\infty \mathbb{P}(Z \geq t) dt \\ &\leq A + \int_A^\infty \exp \left\{ - \left( \frac{t - A}{B} \right)^2 \right\} dt. \\ &= A + \frac{\sqrt{\pi}B}{2} \end{aligned}$$



## Proof of Theorem 17.

Substituting the definitions of  $A$  and  $B$ , we get

$$\mathbb{E}[\text{MSE}(\mathbb{X}\hat{\theta}^{\mathcal{L}})] \leq \|\theta^*\|_1 \sigma \sqrt{\frac{2\log(2d)}{n}} + 2\sqrt{2\pi} \|\theta^*\|_1 \sigma \frac{1}{\sqrt{n}}.$$

Since  $\log(2d) \geq \log 2$ , there exists a numerical constant  $C > 0$  such that

$$\mathbb{E}[\text{MSE}(\mathbb{X}\hat{\theta}^{\mathcal{L}})] \leq C \|\theta^*\|_1 \sigma \sqrt{\frac{\log(2d)}{n}}.$$



## Remark of Theorem 17

- ▶ The regularization parameter ( $\diamond$ ) depends on the confidence level  $\delta$ , which is not the case for the BIC estimator ( $\spadesuit$ ).
- ▶ The rate in Theorem 17 is of order  $\sqrt{(\log d)/n}$  (**slow rate**), which is much slower than the rate of order  $(\log d)/n$  (**fast rate**) for the BIC estimator.
- ▶ Fast rates can be achieved by the computationally efficient Lasso estimator but at the cost of a much stronger condition on the design matrix.

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- ▶ Fast rates can be achieved by the computationally efficient Lasso estimator but at the cost of a much stronger condition on the design matrix.

The BIC estimator uses an  $\ell_0$ -type penalty, which directly penalizes the number of selected variables and therefore exploits the underlying sparsity structure more explicitly than the  $\ell_1$  penalty. Under suitable conditions, this sharper structural adaptation leads to the fast rate  $(\log d)/n$ , whereas the basic Lasso slow-rate analysis only controls the maximal noise-covariate correlation and yields the slower rate  $\sqrt{(\log d)/n}$ .

# Incoherence

## Assumption INC( $k$ )

We say that the design matrix  $\mathbb{X}$  has incoherence  $k$  for some integer  $k > 0$  if

$$\left| \frac{\mathbb{X}^T \mathbb{X}}{n} - I_d \right|_{\infty} \leq \frac{1}{14k}$$

where the  $|A|_{\infty}$  denotes the largest element of  $A$  in absolute value. Equivalently,

1. For all  $j = 1, \dots, d$ ,

$$\left| \frac{\|\mathbb{X}_j\|_2^2}{n} - 1 \right| \leq \frac{1}{14k}.$$

2. For all  $1 \leq i, j \leq d$ ,  $i \neq j$ , we have

$$|\mathbb{X}_i^T \mathbb{X}_j| \leq \frac{1}{14k}.$$

### Remark of Assumption $\text{INC}(k)$

- ▶ Assumption ORT arises as the limiting case of  $\text{INC}(k)$  as  $k \rightarrow \infty$ .
- ▶ While Assumption ORT requires  $d \leq n$ ,  $\text{INC}(k)$  may have  $d \gg n$ .



# Incoherence

## Proposition 2

Let  $\mathbb{X} \in \mathbb{R}^{n \times d}$  be a random matrix with entries  $X_{i,j}, i = 1, \dots, n, j = 1, \dots, d$  that are i.i.d. Rademacher ( $\pm 1$ ) random variables. Then  $\mathbb{X}$  has incoherence  $k$  with probability  $1 - \delta$  as soon as

$$n \geq 392k^2 \log(1/\delta) + 784k^2 \log(d).$$

It implies that there exists matrices that satisfy Assumption  $\text{INC}(k)$  for

$$n \gtrsim k^2 \log(d),$$

for some numerical constant  $C$ .

► **There exists a matrix that satisfies  $\text{INC}(k)$  even for  $d > n$ .**

## Proof of Proposition 2.

Let  $\epsilon_{i,j} \in \{-1, 1\}$  denote the Rademacher random variable that is on the  $i$ th row and  $j$ th column of  $\mathbb{X}$ .

Note first that the  $j$ th diagonal entries of  $\mathbb{X}^T \mathbb{X} / n$  is given by

$$\frac{1}{n} \sum_{i=1}^n \epsilon_{i,j}^2 = 1.$$

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$$\frac{1}{n} \sum_{i=1}^n \epsilon_{i,j}^2 = 1.$$

Moreover, for  $j \neq k$ , the  $(j, k)$ th entry of the  $d \times d$  matrix  $\frac{\mathbb{X}^T \mathbb{X}}{n}$  is given by

$$\frac{1}{n} \sum_{i=1}^n \epsilon_{i,j} \epsilon_{i,k} = \frac{1}{n} \sum_{i=1}^n \xi_i^{(j,k)},$$

where for each pair,  $(j, k)$ ,  $\xi_i^{(j,k)} = \epsilon_{i,j} \epsilon_{i,k}$  so that the random variables  $\xi_1^{(j,k)}, \dots, \xi_n^{(j,k)}$  are iid Rademacher variables. □

## Proof of Proposition 2.

Therefore, we get that for any  $t > 0$ ,

$$\mathbb{P} \left( \left| \frac{\mathbb{X}^T \mathbb{X}}{n} - I_d \right|_{\infty} > 0 \right) = \mathbb{P} \left( \max_{j \neq k} \left| \frac{1}{n} \sum_{i=1}^n \xi_i^{(j,k)} \right| > t \right)$$

## Proof of Proposition 2.

Therefore, we get that for any  $t > 0$ ,

$$\begin{aligned}\mathbb{P}\left(\left|\frac{\mathbb{X}^T \mathbb{X}}{n} - I_d\right|_{\infty} > 0\right) &= \mathbb{P}\left(\max_{j \neq k} \left|\frac{1}{n} \sum_{i=1}^n \xi_i^{(j,k)}\right| > t\right) \\ &\leq \sum_{j \neq k} \mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n \xi_i^{(j,k)}\right| > t\right)\end{aligned}\quad (\text{Union bound})$$

## Proof of Proposition 2.

Therefore, we get that for any  $t > 0$ ,

$$\begin{aligned}\mathbb{P}\left(\left|\frac{\mathbb{X}^T \mathbb{X}}{n} - I_d\right|_{\infty} > 0\right) &= \mathbb{P}\left(\max_{j \neq k} \left|\frac{1}{n} \sum_{i=1}^n \xi_i^{(j,k)}\right| > t\right) \\ &\leq \sum_{j \neq k} \mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n \xi_i^{(j,k)}\right| > t\right) && \text{(Union bound)} \\ &\leq \sum_{j \neq k} 2e^{-\frac{nt^2}{2}} && \text{(Hoeffding's inequality)} \\ &\leq d^2 e^{-\frac{nt^2}{2}},\end{aligned}$$

where we use the fact that  $\xi_i^{(j,k)} \sim \text{subG}(1^2)$ . □

## Proof of Proposition 2.

Taking now  $t = 1/(14k)$  yields

$$\mathbb{P} \left( \left| \frac{\mathbb{X}^T \mathbb{X}}{n} - I_d \right| > \frac{1}{14k} \right) \leq d^2 e^{-\frac{n}{392k^2}} \leq \delta$$

for

$$n \geq 392k^2 \log(1/\delta) + 784k^2 \log(d).$$



# Incoherence

For any  $\theta \in \mathbb{R}^d$  and  $S \subset \{1, \dots, d\}$ , define  $\theta_S$  to be the vector with coordinates

$$\theta_{S,j} = \begin{cases} \theta_j & \text{if } j \in S, \\ 0 & \text{otherwise.} \end{cases}$$

In particular  $\|\theta\|_1 = \|\theta_S\|_1 + \|\theta_{S^c}\|_1$ .



## Lemma 18

Fix a positive integer  $k \leq d$  and assume that  $\mathbb{X}$  satisfies assumption  $\text{INC}(k)$ . Then for any  $S \in \{1, \dots, d\}$  such that  $|S| \leq k$  and any  $\theta \in \mathbb{R}^d$  that satisfies the cone condition

$$\|\theta_{S^c}\|_1 \leq 3\|\theta_S\|_1, \quad (\heartsuit)$$

it holds

$$\|\theta_S\|_2^2 \leq 2 \frac{\|\mathbb{X}\theta\|_2^2}{n}$$

## Proof of Lemma 18.

We have

$$\begin{aligned}\frac{\|\mathbb{X}\theta\|_2^2}{n} &= \frac{1}{n}\|\mathbb{X}\theta_S + \mathbb{X}\theta_{S^c}\|_2^2 = \frac{\|\mathbb{X}\theta_S\|_2^2}{n} + 2\theta_S^T \frac{\mathbb{X}^T \mathbb{X}}{n} \theta_{S^c} + \frac{\|\mathbb{X}\theta_{S^c}\|_2^2}{n}. \\ &\geq \frac{\|\mathbb{X}\theta_S\|_2^2}{n} + 2\theta_S^T \frac{\mathbb{X}^T \mathbb{X}}{n} \theta_{S^c}.\end{aligned}$$

## Proof of Lemma 18.

We have

$$\begin{aligned}\frac{\|\mathbb{X}\theta\|_2^2}{n} &= \frac{1}{n}\|\mathbb{X}\theta_S + \mathbb{X}\theta_{S^c}\|_2^2 = \frac{\|\mathbb{X}\theta_S\|_2^2}{n} + 2\theta_S^T \frac{\mathbb{X}^T \mathbb{X}}{n} \theta_{S^c} + \frac{\|\mathbb{X}\theta_{S^c}\|_2^2}{n}. \\ &\geq \frac{\|\mathbb{X}\theta_S\|_2^2}{n} + 2\theta_S^T \frac{\mathbb{X}^T \mathbb{X}}{n} \theta_{S^c}.\end{aligned}$$

Moreover, from the incoherence condition, we have

$$\begin{aligned}\frac{\|\mathbb{X}\theta_S\|_2^2}{n} &= \theta_S^T \frac{\mathbb{X}^T \mathbb{X}}{n} \theta_{S^c} = \|\theta_S\|_2^2 + \theta_S^T \left( \frac{\mathbb{X}^T \mathbb{X}}{n} - I_d \right) \theta_S \\ &\geq \|\theta_S\|_2^2 - \frac{\|\theta_S\|_1^2}{14k} \quad \textbf{(Check By Yourself)}\end{aligned}$$

□

## Proof of Lemma 18.

By the definition of  $\theta_S$  and  $\theta_{S^c}$ , we have  $\theta_{S,i}\theta_{S^c,i} = 0$ . And then

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$$\left| \theta_S^T \frac{\mathbb{X}^T \mathbb{X}}{n} \theta_{S^c} \right| = \left| \sum_{i=1}^d \sum_{j=1}^d \theta_{S,i} \left( \frac{\mathbb{X}_i^T \mathbb{X}_j}{n} \right) \theta_{S^c,j} \right|$$

## Proof of Lemma 18.

By the definition of  $\theta_S$  and  $\theta_{S^c}$ , we have  $\theta_{S,i}\theta_{S^c,i} = 0$ . And then

$$\begin{aligned}\left| \theta_S^T \frac{\mathbb{X}^T \mathbb{X}}{n} \theta_{S^c} \right| &= \left| \sum_{i=1}^d \sum_{j=1}^d \theta_{S,i} \left( \frac{\mathbb{X}_i^T \mathbb{X}_j}{n} \right) \theta_{S^c,j} \right| \\ &= \sum_{i \neq j} \left| \theta_{S,i} \left( \frac{\mathbb{X}_i^T \mathbb{X}_j}{n} \right) \theta_{S^c,j} \right| \\ &\leq \frac{1}{14k} \sum_{i \neq j} |\theta_{S,i}| \cdot |\theta_{S^c,j}| \quad \textbf{(INC}(k))\end{aligned}$$

## Proof of Lemma 18.

By the definition of  $\theta_S$  and  $\theta_{S^c}$ , we have  $\theta_{S,i}\theta_{S^c,i} = 0$ . And then

$$\begin{aligned}\left| \theta_S^T \frac{\mathbb{X}^T \mathbb{X}}{n} \theta_{S^c} \right| &= \left| \sum_{i=1}^d \sum_{j=1}^d \theta_{S,i} \left( \frac{\mathbb{X}_i^T \mathbb{X}_j}{n} \right) \theta_{S^c,j} \right| \\ &= \sum_{i \neq j} \left| \theta_{S,i} \left( \frac{\mathbb{X}_i^T \mathbb{X}_j}{n} \right) \theta_{S^c,j} \right| \\ &\leq \frac{1}{14k} \sum_{i \neq j} |\theta_{S,i}| \cdot |\theta_{S^c,j}| \quad \textbf{(INC}(k)) \\ &\leq \frac{1}{14k} \|\theta_S\|_1 \|\theta_{S^c}\|_1\end{aligned}$$

## Proof of Lemma 18.

By the definition of  $\theta_S$  and  $\theta_{S^c}$ , we have  $\theta_{S,i}\theta_{S^c,i} = 0$ . And then

$$\begin{aligned}\left| \theta_S^T \frac{\mathbb{X}^T \mathbb{X}}{n} \theta_{S^c} \right| &= \left| \sum_{i=1}^d \sum_{j=1}^d \theta_{S,i} \left( \frac{\mathbb{X}_i^T \mathbb{X}_j}{n} \right) \theta_{S^c,j} \right| \\ &= \sum_{i \neq j} \left| \theta_{S,i} \left( \frac{\mathbb{X}_i^T \mathbb{X}_j}{n} \right) \theta_{S^c,j} \right| \\ &\leq \frac{1}{14k} \sum_{i \neq j} |\theta_{S,i}| \cdot |\theta_{S^c,j}| \quad \textbf{(INC}(k)) \\ &\leq \frac{1}{14k} \|\theta_S\|_1 \|\theta_{S^c}\|_1 \\ &\leq \frac{3}{14k} \|\theta_S\|_1^2 \quad \textbf{(Lemma's condition)}\end{aligned}$$

□



## Proof of Lemma 18.

Putting together,

$$\begin{aligned}\frac{\|\mathbb{X}\theta\|_2^2}{n} &\geq \frac{\|\mathbb{X}\theta_S\|_2^2}{n} + 2\theta_S^T \frac{\mathbb{X}^T \mathbb{X}}{n} \theta_{S^c} \\ &\geq \|\theta_S\|_2^2 - \frac{\|\theta_S\|_1^2}{14k} - \frac{6}{14k} \|\theta_S\|_1^2 \\ &= \|\theta_S\|_2^2 - \frac{1}{2k} \|\theta_S\|_1^2.\end{aligned}$$

## Proof of Lemma 18.

Putting together,

$$\begin{aligned}\frac{\|\mathbb{X}\theta\|_2^2}{n} &\geq \frac{\|\mathbb{X}\theta_S\|_2^2}{n} + 2\theta_S^T \frac{\mathbb{X}^T \mathbb{X}}{n} \theta_{S^c} \\ &\geq \|\theta_S\|_2^2 - \frac{\|\theta_S\|_1^2}{14k} - \frac{6}{14k} \|\theta_S\|_1^2 \\ &= \|\theta_S\|_2^2 - \frac{1}{2k} \|\theta_S\|_1^2.\end{aligned}$$

Since  $\|\theta_S\|_1^2 \leq |S| \|\theta_S\|_2^2$  and  $|S| \leq k$ , we complete the proof. □

## Theorem 19

Fix  $n \geq 2$ . Assume that the (Linear Model) holds where  $\epsilon \sim \text{subG}_n(\sigma^2)$ . Moreover, assume that  $\|\theta^*\|_0 \leq k$  and that  $\mathbb{X}$  satisfies assumption  $\text{INC}(k)$ . Then the Lasso estimator  $\hat{\theta}^{\mathcal{L}}$  with regularization parameter defined by

$$2\tau = 8\sigma\sqrt{\frac{\log(2d)}{n}} + 8\sigma\sqrt{\frac{\log(1/\delta)}{n}}$$

satisfies

$$\text{MSE}(\mathbb{X}\hat{\theta}^{\mathcal{L}}) = \frac{1}{n}\|\mathbb{X}\hat{\theta}^{\mathcal{L}} - \mathbb{X}\theta^*\|_2^2 \lesssim k\sigma^2\frac{\log(2d/\delta)}{n}$$

and

$$\|\hat{\theta}^{\mathcal{L}} - \theta^*\|_1 \lesssim k\sigma\sqrt{\frac{\log(2d/\delta)}{n}}$$

with probability at least  $1 - \delta$ . Moreover,

$$\mathbb{E}[\text{MSE}(\mathbb{X}\hat{\theta}^{\mathcal{L}})] \lesssim k\sigma^2\frac{\log(2d)}{n}, \quad \text{and} \quad \mathbb{E}[\|\hat{\theta}^{\mathcal{L}} - \theta^*\|_1] \lesssim k\sigma\sqrt{\frac{\log(2d/\delta)}{n}}.$$

## Proof of Theorem 19.

From the definition of  $\hat{\theta}^{\mathcal{L}}$ ,

$$\frac{1}{n} \|Y - \mathbb{X} \hat{\theta}^{\mathcal{L}}\|_2^2 + 2\tau \|\hat{\theta}^{\mathcal{L}}\|_1 \leq \frac{1}{n} \|Y - \mathbb{X} \theta^*\|_2^2 + 2\tau \|\theta^*\|_1$$

## Proof of Theorem 19.

From the definition of  $\hat{\theta}^{\mathcal{L}}$ ,

$$\frac{1}{n}\|Y - \mathbb{X}\hat{\theta}^{\mathcal{L}}\|_2^2 + 2\tau\|\hat{\theta}^{\mathcal{L}}\|_1 \leq \frac{1}{n}\|Y - \mathbb{X}\theta^*\|_2^2 + 2\tau\|\theta^*\|_1$$

Multiplying both sides by  $n$  and similar to the proof of Theorem 17, we have

$$\|\mathbb{X}\hat{\theta}^{\mathcal{L}} - \mathbb{X}\theta^*\|_2^2 \leq 2\epsilon^T \mathbb{X}(\hat{\theta}^{\mathcal{L}} - \theta^*) + 2\tau n\|\theta^*\|_1 - 2\tau n\|\hat{\theta}^{\mathcal{L}}\|_1$$

## Proof of Theorem 19.

From the definition of  $\hat{\theta}^{\mathcal{L}}$ ,

$$\frac{1}{n}\|Y - \mathbb{X}\hat{\theta}^{\mathcal{L}}\|_2^2 + 2\tau\|\hat{\theta}^{\mathcal{L}}\|_1 \leq \frac{1}{n}\|Y - \mathbb{X}\theta^*\|_2^2 + 2\tau\|\theta^*\|_1$$

Multiplying both sides by  $n$  and similar to the proof of Theorem 17, we have

$$\|\mathbb{X}\hat{\theta}^{\mathcal{L}} - \mathbb{X}\theta^*\|_2^2 \leq 2\epsilon^T \mathbb{X}(\hat{\theta}^{\mathcal{L}} - \theta^*) + 2\tau n\|\theta^*\|_1 - 2\tau n\|\hat{\theta}^{\mathcal{L}}\|_1$$

And adding both sides  $n\tau\|\hat{\theta}^{\mathcal{L}} - \theta^*\|_1$  yields

$$\|\mathbb{X}\hat{\theta}^{\mathcal{L}} - \mathbb{X}\theta^*\|_2^2 + n\tau\|\hat{\theta}^{\mathcal{L}} - \theta^*\|_1 \leq 2\epsilon^T \mathbb{X}(\hat{\theta}^{\mathcal{L}} - \theta^*) + n\tau\|\hat{\theta}^{\mathcal{L}} - \theta^*\|_1 + 2\tau n\|\theta^*\|_1 - 2\tau n\|\hat{\theta}^{\mathcal{L}}\|_1$$

□

## Proof of Theorem 19.

Applying Hölder's inequality, we get

$$\epsilon^T \mathbb{X}(\hat{\theta}^{\mathcal{L}} - \theta^*) \leq \|\epsilon^T \mathbb{X}\|_{\infty} \|\hat{\theta}^{\mathcal{L}} - \theta^*\|_1.$$

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By the assumption  $\text{INC}(k)$ , we have  $\|\mathbb{X}_j\|_2^2 \leq (\frac{1}{14k} + 1)n \leq 2n$ . And hence,  $\epsilon^T \mathbb{X}_j \sim \text{subG}(2n\sigma^2)$ .



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Following the same steps in the proof of Theorem 19, we get

$$\epsilon^T \mathbb{X}(\hat{\theta}^{\mathcal{L}} - \theta^*) \leq \frac{n\tau}{2} \|\hat{\theta}^{\mathcal{L}} - \theta^*\|_1.$$

with probability at least  $1 - \delta$ .

□

## Proof of Theorem 19.

Therefore, taking  $S = \text{supp}(\theta^*)$  to be the support of  $\theta^*$ , we get

$$\begin{aligned}\|\mathbb{X}\hat{\theta}^{\mathcal{L}} - \mathbb{X}\theta^*\|_2^2 + n\tau\|\hat{\theta}^{\mathcal{L}} - \theta^*\|_1 &\leq 2n\tau\|\hat{\theta}^{\mathcal{L}} - \theta^*\|_1 + 2n\tau\|\theta^*\|_1 - 2n\tau\|\hat{\theta}^{\mathcal{L}}\|_1 \\ &= 2n\tau\|\hat{\theta}_S^{\mathcal{L}} + \hat{\theta}_{S^c}^{\mathcal{L}} - \theta^*\|_1 + 2n\tau\|\theta^*\|_1 - 2n\tau\|\hat{\theta}_S^{\mathcal{L}} + \hat{\theta}_{S^c}^{\mathcal{L}}\|_1\end{aligned}$$

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Therefore, taking  $S = \text{supp}(\theta^*)$  to be the support of  $\theta^*$ , we get

$$\begin{aligned}\|\mathbb{X}\hat{\theta}^{\mathcal{L}} - \mathbb{X}\theta^*\|_2^2 + n\tau\|\hat{\theta}^{\mathcal{L}} - \theta^*\|_1 &\leq 2n\tau\|\hat{\theta}^{\mathcal{L}} - \theta^*\|_1 + 2n\tau\|\theta^*\|_1 - 2n\tau\|\hat{\theta}^{\mathcal{L}}\|_1 \\ &= 2n\tau\|\hat{\theta}_S^{\mathcal{L}} + \hat{\theta}_{S^c}^{\mathcal{L}} - \theta^*\|_1 + 2n\tau\|\theta^*\|_1 - 2n\tau\|\hat{\theta}_S^{\mathcal{L}} + \hat{\theta}_{S^c}^{\mathcal{L}}\|_1 \\ &= 2n\tau\|\hat{\theta}_S^{\mathcal{L}} - \theta^*\|_1 + 2n\tau\|\theta^*\|_1 - 2n\tau\|\hat{\theta}_S^{\mathcal{L}}\|_1\end{aligned}$$

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Therefore, taking  $S = \text{supp}(\theta^*)$  to be the support of  $\theta^*$ , we get

$$\begin{aligned}\|\mathbb{X}\hat{\theta}^{\mathcal{L}} - \mathbb{X}\theta^*\|_2^2 + n\tau\|\hat{\theta}^{\mathcal{L}} - \theta^*\|_1 &\leq 2n\tau\|\hat{\theta}^{\mathcal{L}} - \theta^*\|_1 + 2n\tau\|\theta^*\|_1 - 2n\tau\|\hat{\theta}^{\mathcal{L}}\|_1 \\ &= 2n\tau\|\hat{\theta}_S^{\mathcal{L}} + \hat{\theta}_{S^c}^{\mathcal{L}} - \theta^*\|_1 + 2n\tau\|\theta^*\|_1 - 2n\tau\|\hat{\theta}_S^{\mathcal{L}} + \hat{\theta}_{S^c}^{\mathcal{L}}\|_1 \\ &= 2n\tau\|\hat{\theta}_S^{\mathcal{L}} - \theta^*\|_1 + 2n\tau\|\theta^*\|_1 - 2n\tau\|\hat{\theta}_S^{\mathcal{L}}\|_1 \\ &\leq 4n\tau\|\hat{\theta}_S^{\mathcal{L}} - \theta^*\|_1 \quad (\text{Triangle Inequality})\end{aligned}\tag{†}$$

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Therefore, taking  $S = \text{supp}(\theta^*)$  to be the support of  $\theta^*$ , we get

$$\begin{aligned}\|\mathbb{X}\hat{\theta}^{\mathcal{L}} - \mathbb{X}\theta^*\|_2^2 + n\tau\|\hat{\theta}^{\mathcal{L}} - \theta^*\|_1 &\leq 2n\tau\|\hat{\theta}^{\mathcal{L}} - \theta^*\|_1 + 2n\tau\|\theta^*\|_1 - 2n\tau\|\hat{\theta}^{\mathcal{L}}\|_1 \\ &= 2n\tau\|\hat{\theta}_S^{\mathcal{L}} + \hat{\theta}_{S^c}^{\mathcal{L}} - \theta^*\|_1 + 2n\tau\|\theta^*\|_1 - 2n\tau\|\hat{\theta}_S^{\mathcal{L}} + \hat{\theta}_{S^c}^{\mathcal{L}}\|_1 \\ &= 2n\tau\|\hat{\theta}_S^{\mathcal{L}} - \theta^*\|_1 + 2n\tau\|\theta^*\|_1 - 2n\tau\|\hat{\theta}_S^{\mathcal{L}}\|_1 \\ &\leq 4n\tau\|\hat{\theta}_S^{\mathcal{L}} - \theta^*\|_1 \quad (\text{Triangle Inequality})\end{aligned}\tag{†}$$

Hence

$$\|\hat{\theta}^{\mathcal{L}} - \theta^*\|_1 \leq 4\|\hat{\theta}_S^{\mathcal{L}} - \theta^*\|_1 = 4\|\hat{\theta}_S^{\mathcal{L}} - \theta_S^*\|_1.$$

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Therefore, taking  $S = \text{supp}(\theta^*)$  to be the support of  $\theta^*$ , we get

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Hence

$$\|\hat{\theta}^{\mathcal{L}} - \theta^*\|_1 \leq 4\|\hat{\theta}_S^{\mathcal{L}} - \theta^*\|_1 = 4\|\hat{\theta}_S^{\mathcal{L}} - \theta_S^*\|_1.$$

Subtracting both sides by  $\|\hat{\theta}_S^{\mathcal{L}} - \theta_S^*\|_1$  yields

$$\|\hat{\theta}_{S^c}^{\mathcal{L}} - \theta_{S^c}^*\|_1 \leq 3\|\hat{\theta}_S^{\mathcal{L}} - \theta_S^*\|_1.$$

□

## Proof of Theorem 19.

So that  $\theta = \hat{\theta}^{\mathcal{L}} - \theta^*$  satisfies the cone condition ( $\heartsuit$ ). Using the Cauchy-Schwartz inequality and Lemma 18 respectively, we get since  $|S| \leq k$ ,

$$\|\hat{\theta}^{\mathcal{L}} - \theta^*\|_1 \leq \sqrt{S} \|\hat{\theta}_S^{\mathcal{L}} - \theta^*\|_2 \leq \sqrt{\frac{2k}{n}} \|\mathbb{X} \hat{\theta}^{\mathcal{L}} - \mathbb{X} \theta^*\|_2$$

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Combing this result with ( $\dagger$ ), we find

